203738: porphyritic metasyenite, Lake Carey

(Yilgarn Craton granites, Kurnalpi Terrane, Eastern Goldfields Superterrane, Yilgarn Craton)

Location and sampling

Laverton (SH 51-2), Laverton (3340)
MGA Zone 51, 436769E 6797459N

Sampled on 2 September 2020

This sample was collected from the 341.0 – 343.0 m depth interval of drillcore LCDD0076, drilled in 2017 by GSM Mining Company Pty Ltd at their Northern Fleet prospect. The drillhole is located on Mount Weld Station, in the northeastern part of Lake Carey, about 20 km southeast of Mount Margaret, 13.1 km north-northwest of Sunrise Dam, and 9.3 km west-southwest of Brumby Well.

Geological context

The unit sampled is a porphyritic metasyenite (Fig. 1) within the Laverton greenstone belt of the Kurnalpi Terrane (Stewart, 2001; Standing, 2008). The unit consists of multiple intrusions, mainly metamorphosed syenite and monzonite, that intruded metasedimentary and minor felsic metavolcanic rocks in the Lake Carey domain (Standing, 2008). A felsic volcanic rock about 0.9 km to the southwest in drillcore LCDD0013 yielded an igneous crystallization age of 2696 ± 6 Ma (GSWA 203739, Wingate et al., 2022). A porphyritic metasyenite at the Wallaby mine, about 11.1 km to the north-northwest, indicated an igneous crystallization age of 2688 ± 27 Ma (GSWA 203740, Wingate et al., 2022b). Detrital zircons in a sandstone horizon within conglomerate at the Wallaby mine, about 11.8 km to the north-northwest, yielded a conservative maximum depositional age of 2673 ± 5 Ma (sample E319, Kositcin et al., 2008). An andesite-derived sandstone at the Sunrise Dam mine, about 15.8 km to the south-southeast, yielded a conservative maximum depositional age of 2715 ± 5 Ma (sample E324, Kositcin et al., 2008). Detrital zircons in a polymictic breccia unit at the Granny Smith mine, about 18.4 km to the north-northeast, indicated a youngest detrital age component at 2663 ± 5 Ma (sample E401, Kositcin et al., 2008). A feldspar–quartz-porphyritic rock at Ida Hill, about 27 km to the north-northeast, provided an igneous crystallization age of c. 2660 Ma (sample E258, Kositcin et al., 2008).

Petrographic description

The sample is a medium-grained, porphyritic metasyenite (Fig. 1). The porphyritic texture is mainly defined by sparse, subhedral to euhedral, zoned K-feldspar phenocrysts and rare plagioclase phenocrysts. Smaller phenocrysts of elongate to equant biotite are partly altered to chlorite. Some biotite contains numerous, very fine acicular inclusions of rutile and likely represents former amphibole phenocrysts. The groundmass is dominated by granular-textured anhedral to subhedral feldspar laths (K-feldspar and minor plagioclase), and minor interstitial quartz. Minor acicular to equant apatite crystals form aggregates associated with biotite. Accessory minerals include pyrite and zircon. Aggregates and discontinuous stringers of carbonate minerals are associated with fine-grained, irregular biotite and feldspars are partially altered to sericite.

Zircon morphology

Zircons isolated from this sample are colourless to dark brown and anhedral to subhedral. The crystals are up to 200 μm long, and mostly equant, with aspect ratios less than 2:1. In cathodoluminescence (CL) images, concentric zoning is ubiquitous, and some zircons exhibit high-U, metamict outer zones. A CL image of all zircons is shown in Figure 2.

Analytical details

This sample was analysed over three sessions, on 9–10, 15–16, and 22–23 October 2020, using SHRIMP-B. Analyses 1.1 to 6.1 (spot numbers 1–6) were obtained during the first session, together with 13 analyses of the M257 standard, of which 12 analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.24% (1σ). Isotopic mass fractionation of $^{207}Pb/^{206}Pb$ ratios during the first session was corrected by reference to the OGC1 standard; measured ratios were increased by 0.39%. Analyses 7.1 to 4.2 (spot numbers 7–17) were obtained during the second session, together with 16 analyses of the M257 standard, of which 14 analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.28% (1σ). Analyses 11.2 to 8.2 (spot numbers 18–29) were obtained during the third session, together with 12 analyses of the M257 standard, of which 11 analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.31% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Twenty-nine analyses were obtained from 19 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

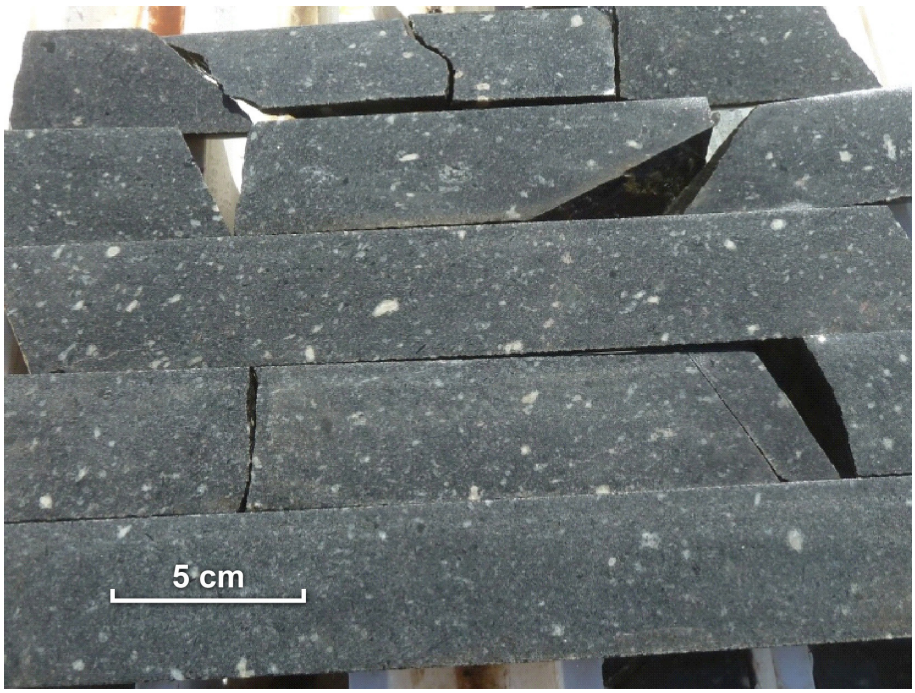

Figure 1. Drillcore image of sample 203738: porphyritic metasyenite, Lake Carey

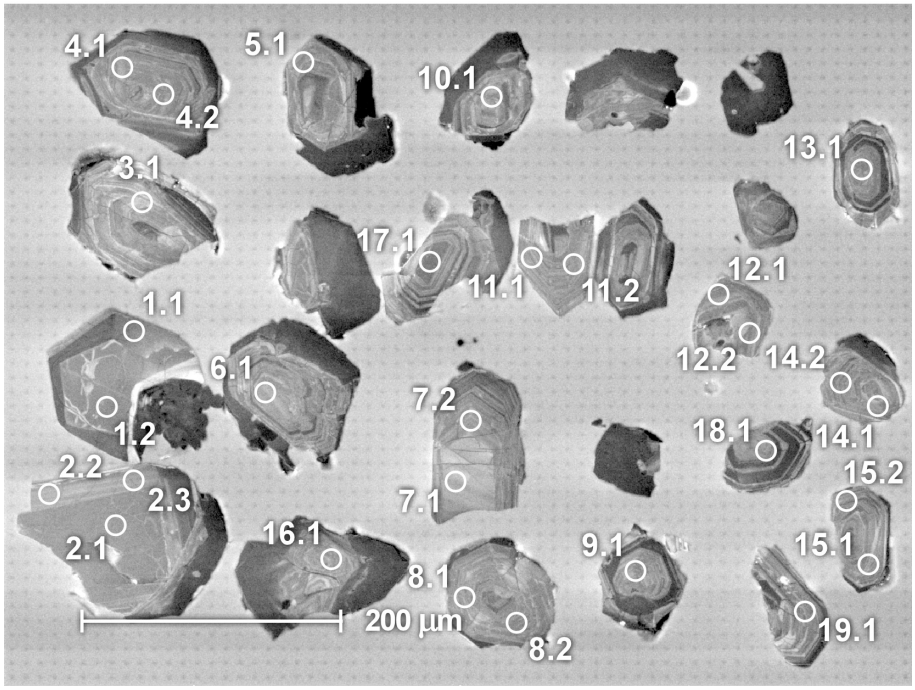

Figure 2. Cathodoluminescence image of all zircons from sample 203738: porphyritic metasyenite, Lake Carey. Numbered circles indicate the approximate locations of analysis sites

Table 1. Ion microprobe analytical results for zircons from sample 203738: porphyritic syenite, Lake Carey

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$	± 1σ	$^{207}Pb/^{206}Pb$	± 1σ	$^{238}U/^{206}Pb^*$	± 1σ	$^{207}Pb^*/^{206}Pb^*$	± 1σ	$^{238}U/^{206}Pb^*$ date (Ma)	±1σ	$^{207}Pb^*/^{206}Pb^*$ date (Ma)	±1σ	Disc. (%)
I	13	13.1	313	343	1.10	0.018	2.018	0.024	0.17939	0.00115	2.019	0.024	0.17923	0.00116	2594	26	2646	11	2.0
I	1	1.1	452	667	1.48	0.025	2.028	0.020	0.17976	0.00148	2.029	0.020	0.17954	0.00149	2583	22	2649	14	2.5
I	19	12.2	281	300	1.07	0.055	2.007	0.024	0.18073	0.00115	2.008	0.025	0.18024	0.00118	2605	27	2655	11	1.9
I	28	7.2	192	114	0.60	0.000	2.017	0.028	0.18107	0.00145	2.017	0.028	0.18107	0.00145	2595	30	2663	13	2.5
I	2	2.1	243	319	1.31	0.000	2.060	0.048	0.18131	0.00136	2.060	0.048	0.18131	0.00136	2551	49	2665	12	4.3
I	4	4.1	262	321	1.23	-0.032	2.038	0.025	0.18119	0.00133	2.037	0.025	0.18148	0.00134	2575	26	2666	12	3.4
I	7	7.1	156	91	0.58	-0.126	2.056	0.030	0.18103	0.00150	2.053	0.030	0.18216	0.00160	2558	31	2673	15	4.3
I	16	2.2	141	132	0.94	0.086	2.003	0.032	0.18317	0.00176	2.005	0.032	0.18240	0.00185	2609	35	2675	17	2.5
I	15	1.2	335	395	1.18	-0.049	2.047	0.024	0.18236	0.00110	2.046	0.024	0.18279	0.00112	2565	25	2678	10	4.2
P	9	9.1	197	186	0.94	0.024	2.144	0.028	0.17284	0.00127	2.144	0.028	0.17262	0.00129	2468	27	2583	12	4.5
P	17	4.2	275	269	0.98	0.042	2.147	0.027	0.17326	0.00120	2.148	0.027	0.17288	0.00123	2464	26	2586	12	4.7
P	24	17.1	728	1129	1.55	0.109	2.585	0.047	0.13964	0.00156	2.588	0.047	0.13868	0.00159	2106	33	2211	20	4.7
P	20	14.2	570	667	1.17	0.185	2.509	0.068	0.14385	0.00237	2.514	0.068	0.14221	0.00242	2159	50	2254	29	4.2
D	22	15.2	493	550	1.12	0.129	3.184	0.087	0.14918	0.00186	3.188	0.087	0.14804	0.00190	1759	42	2323	22	24.3
D	21	15.1	417	580	1.39	0.156	2.428	0.025	0.16358	0.00144	2.431	0.025	0.16218	0.00149	2221	20	2479	16	10.4
D	11	11.1	416	343	0.83	0.108	2.232	0.024	0.16744	0.00094	2.234	0.024	0.16648	0.00100	2385	22	2523	10	5.5
D	5	5.1	330	385	1.17	0.051	2.328	0.037	0.16744	0.00119	2.329	0.037	0.16699	0.00121	2303	31	2528	12	8.9
D	29	8.2	367	312	0.85	0.057	2.262	0.027	0.17204	0.00115	2.263	0.027	0.17153	0.00119	2359	24	2573	12	8.3
D	18	11.2	362	319	0.88	0.200	2.314	0.027	0.17530	0.00114	2.319	0.027	0.17351	0.00126	2311	23	2592	12	10.8
D	25	18.1	467	520	1.11	0.167	2.168	0.024	0.17585	0.00097	2.171	0.024	0.17436	0.00105	2442	23	2600	10	6.1
D	3	3.1	171	93	0.54	0.048	2.164	0.027	0.17566	0.00149	2.165	0.027	0.17522	0.00152	2448	26	2608	14	6.1
D	23	16.1	274	269	0.98	-0.037	2.200	0.026	0.17554	0.00113	2.199	0.026	0.17587	0.00116	2416	24	2614	11	7.6
D	8	8.1	360	347	0.96	0.056	2.206	0.024	0.17742	0.00099	2.207	0.024	0.17692	0.00102	2409	22	2624	10	8.2
D	6	6.1	337	330	0.98	0.000	2.219	0.024	0.17705	0.00125	2.219	0.024	0.17705	0.00125	2398	21	2625	12	8.7
D	14	14.1	259	203	0.78	0.181	2.390	0.030	0.17938	0.00135	2.394	0.030	0.17777	0.00148	2250	24	2632	14	14.5
D	10	10.1	218	176	0.81	0.056	2.236	0.029	0.17873	0.00140	2.237	0.029	0.17824	0.00145	2382	27	2637	13	9.7
D	12	12.1	237	224	0.95	0.135	2.174	0.028	0.18042	0.00140	2.177	0.028	0.17922	0.00150	2437	27	2646	14	7.9
D	26	19.1	315	373	1.18	0.038	2.213	0.039	0.17959	0.00119	2.214	0.039	0.17925	0.00122	2403	35	2646	11	9.2
D	27	2.3	146	135	0.93	0.036	2.097	0.030	0.18114	0.00161	2.098	0.030	0.18082	0.00164	2513	31	2660	15	5.5

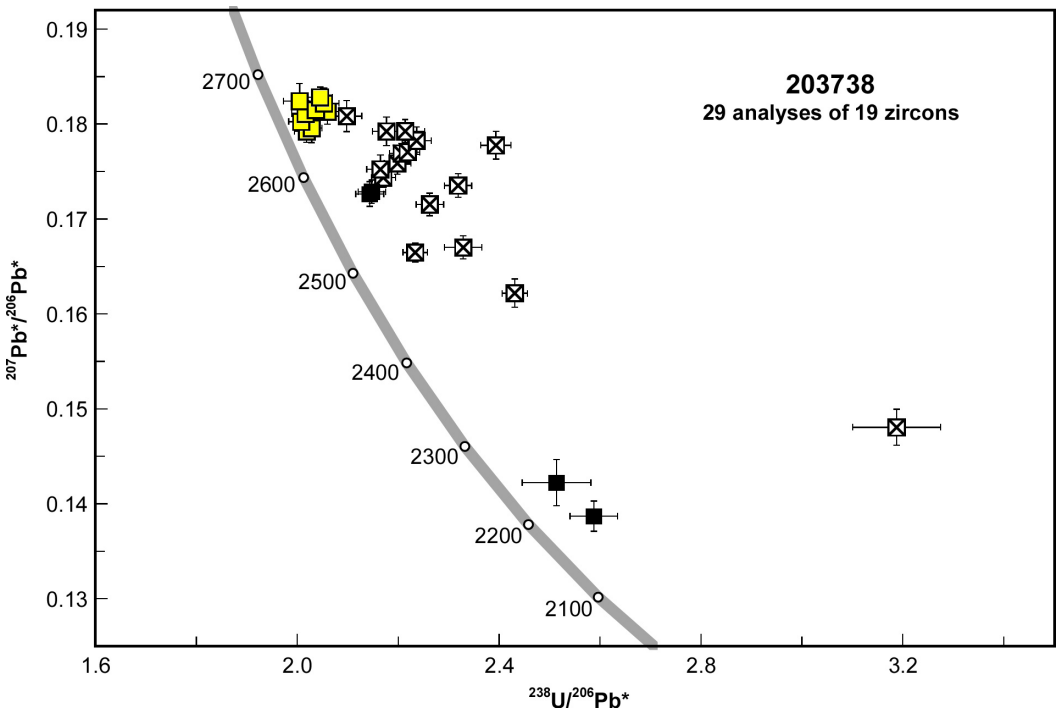

Figure 3. U–Pb analytical data for sample 203738: porphyritic metasyenite, Lake Carey. Yellow squares indicate Group I (magmatic zircons); black squares indicate Group P (radiogenic-Pb loss); crossed squares indicate Group D (discordance >5%)

Interpretation

The analyses are concordant to strongly discordant (Fig. 3). Sixteen analyses are >5% discordant. The dates obtained from these 16 analyses (Group D; Table 1) are unreliable, and are considered not to be geologically significant. The remaining 13 analyses can be divided into two groups, based on their $^{207}Pb^*/^{206}Pb^*$ ratios.

Group I comprises nine analyses of six zircons (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2663 ± 8 Ma (MSWD = 0.94).

Group P comprises four analyses of four zircons (Table 1), which yield $^{207}Pb^*/^{206}Pb^*$ dates of 2586–2211 Ma.

The date of 2663 ± 8 Ma for the nine analyses in Group I is interpreted as a minimum age for igneous crystallization, because the average composition is about 3.1% discordant. The dates of 2586–2211 Ma for the four analyses in Group P are interpreted to reflect loss of radiogenic Pb.

References

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Recommended reference for this publication

Wingate, MTD, Fielding, IOH, Lu, Y, Cassidy, KF and Smithies, RH 2022, 203738: porphyritic metasyenite, Lake Carey; Geochronology Record 1881: Geological Survey of Western Australia, 4p.

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